

Machine Learning

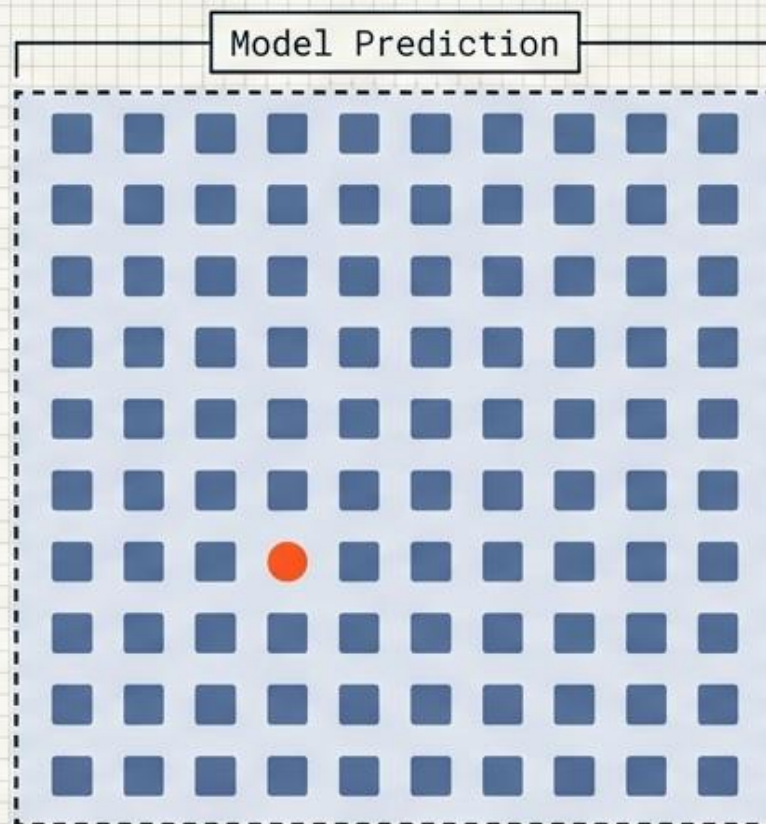
Week 10 Lecture

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Contents

- Revisit:
 - Why Accuracy fails?
 - Confusion matrix
 - Loss function Visualization
- SMOTE
- Cost Sensitive

In imbalanced systems, high accuracy often masks complete model failure



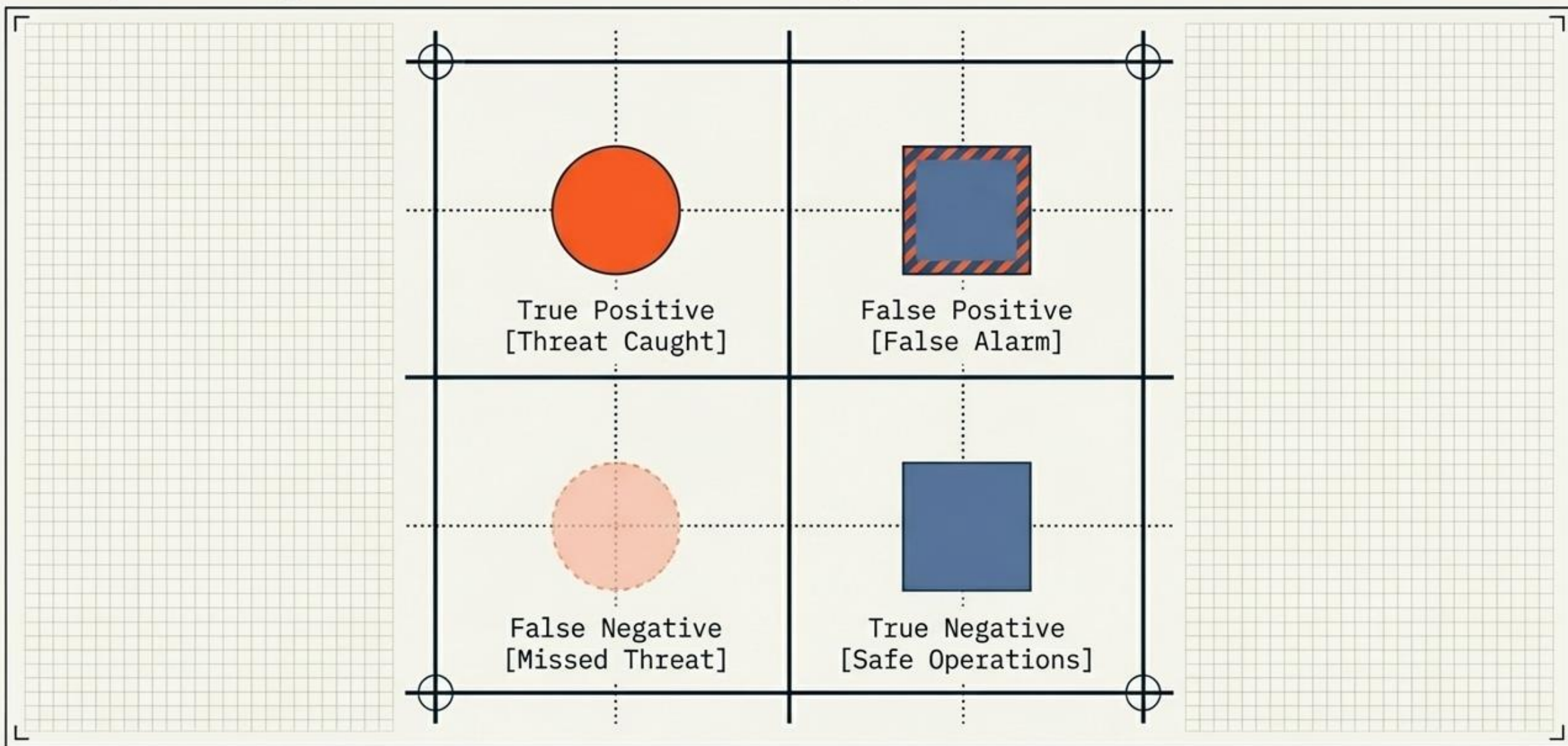
[STATUS: 99% ACCURATE]



[RESULT: 0% USEFUL]

When the majority class dominates, a model can achieve perfect scores simply by predicting that the anomaly never happens.

Every meaningful evaluation metric originates from the Confusion Matrix



☞ This matrix is the fundamental source of truth. It tracks exact failure modes rather than masking them behind aggregate percentages. ☞

Precision and Recall

Precision

		Real Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

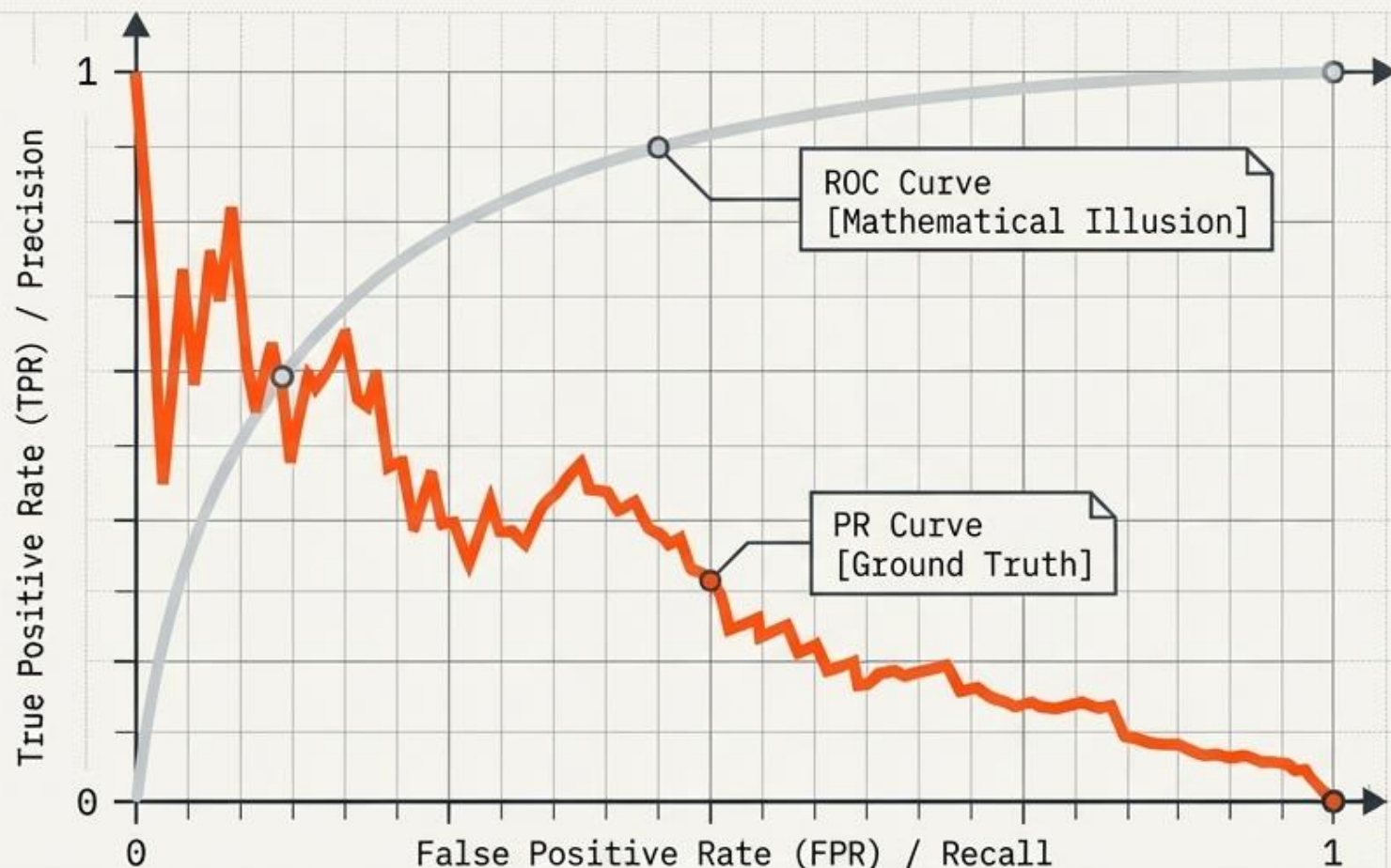
Zeyu, 2021

Recall

		Real Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

Zeyu, 2021

ROC curves inflate perceived performance by masking minority class failures



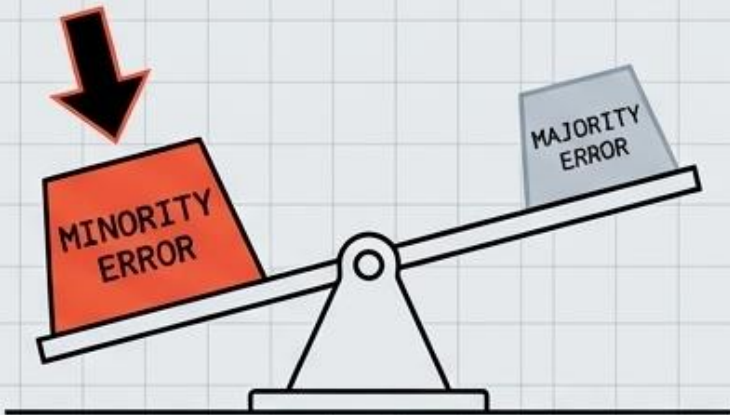
Warning: ROC measures True Positive Rate vs. False Positive Rate.

In imbalanced data, the massive number of True Negatives drags down the FPR, creating the mathematical illusion of success.

Takeaway: Switch to Precision-Recall Curves for severe class imbalance.

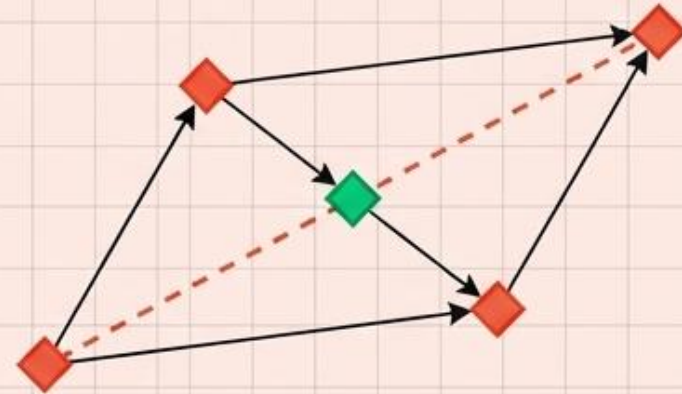
The Two Paradigms of Imbalanced Learning

Algorithmic Modification (Cost-Sensitive Learning)



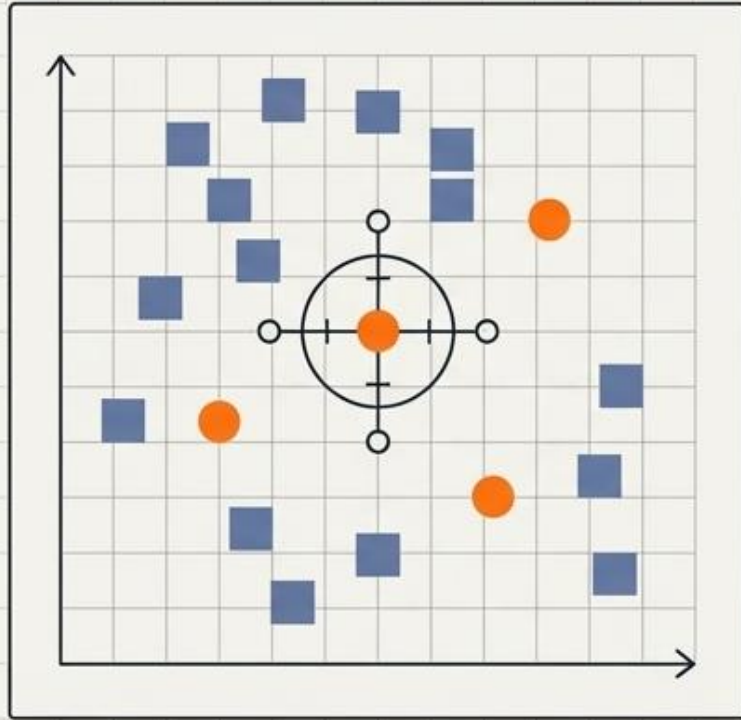
- **Mechanism:** Alters the model's loss function to penalize minority class errors more heavily.
- **Primary Tool:** Cost-Sensitive XGBoost (`scale_pos_weight`).
- **Advantage:** Computationally efficient; no alteration of the underlying dataset.

Data Modification (Synthetic Oversampling)

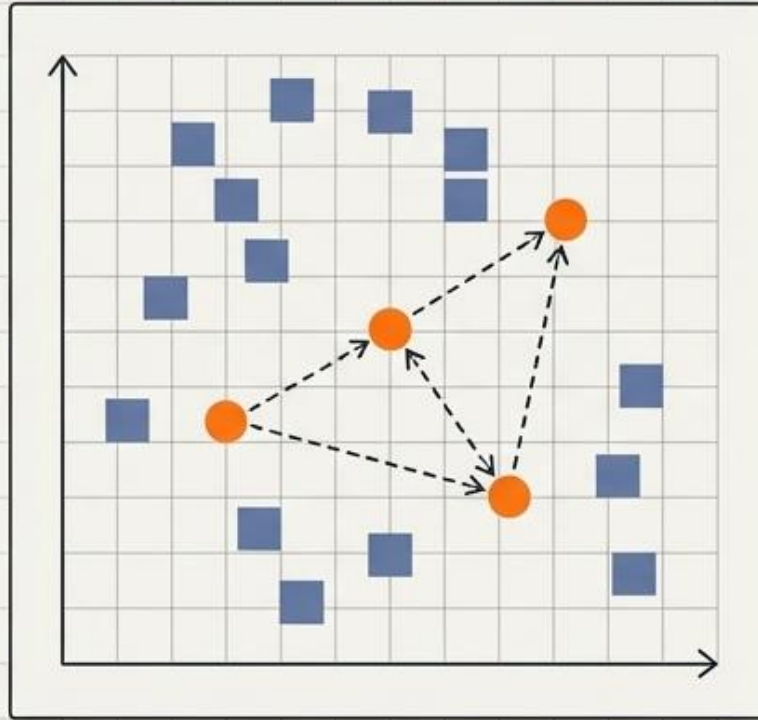


- **Mechanism:** Alters the training dataset prior to modeling by synthesizing new minority instances.
- **Primary Tool:** SMOTE and its advanced variants (Borderline, ENN, SMOTE-ENC).
- **Advantage:** Model-agnostic; physically defines better decision boundaries in the feature space.

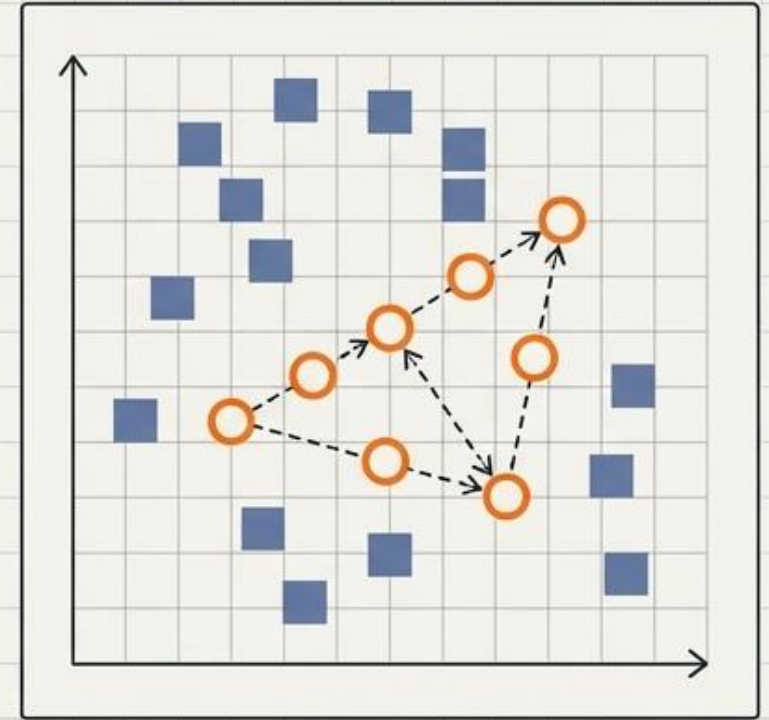
SMOTE mathematically synthesizes new data to balance the feature space



1. Identify



2. Connect

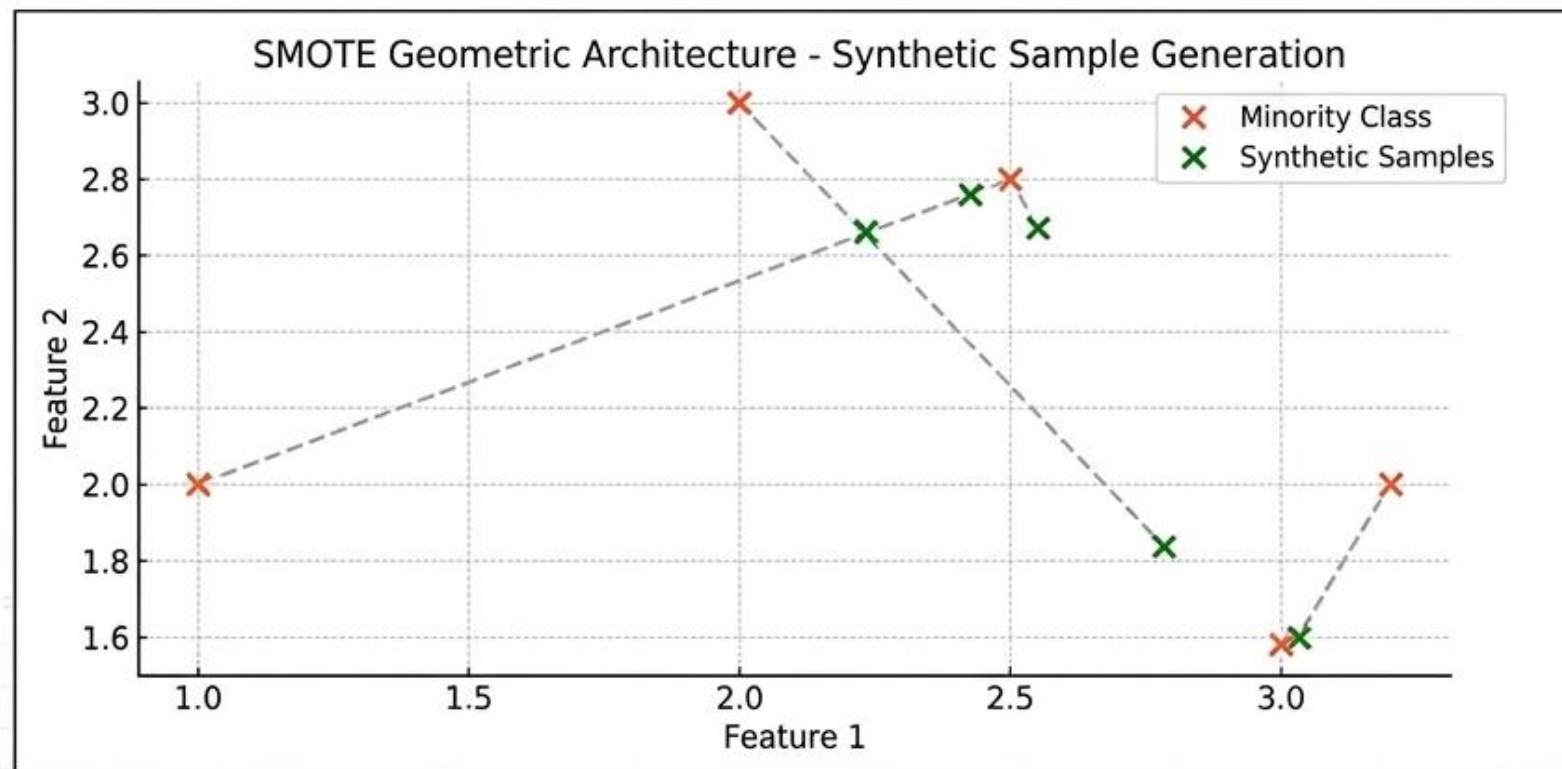


3. Synthesize

Instead of simply duplicating data (which risks overfitting), the Synthetic Minority Oversampling Technique mathematically interpolates new samples along the vectors of existing nearest neighbors.

Data Modification: The Mechanics of SMOTE

The Synthetic Minority Over-sampling Technique (SMOTE) escapes the overfitting trap of random duplication by interpolating entirely new data points.



The Formula Breakdown

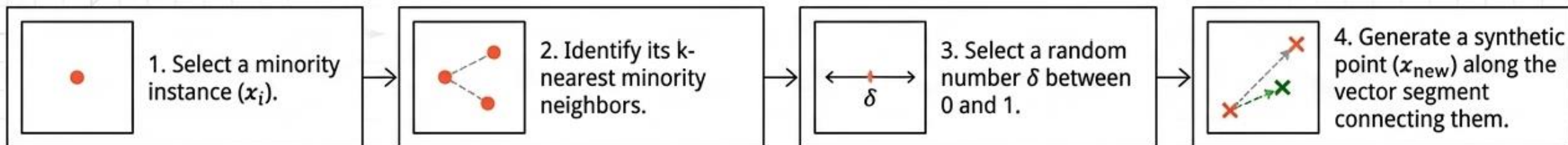
$$x_{new} = x_i + \delta * (x_{neighbor} - x_i)$$

x_{new} : New synthetic data point

x_i : Selected minority instance

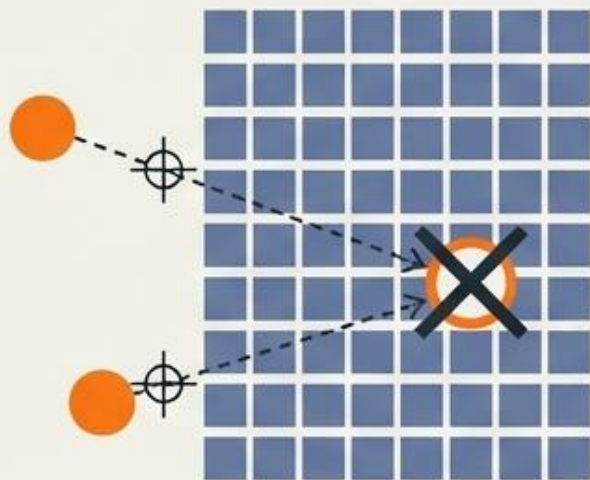
δ : Random number between 0 and 1

$x_{neighbor}$: Nearest minority neighbor



Careless application of SMOTE introduces critical pipeline vulnerabilities

Unrealistic Data



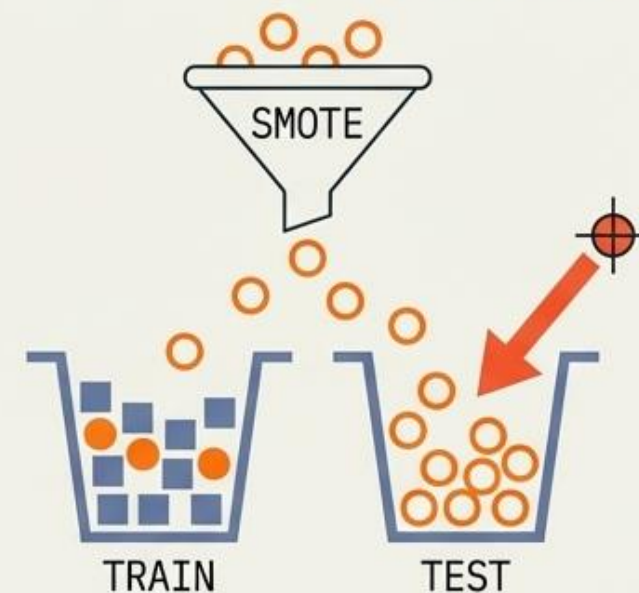
Math interpolation through noisy space generates impossible samples.

Categorical Failure



Geometric distance algorithms fail fundamentally on non-continuous data.

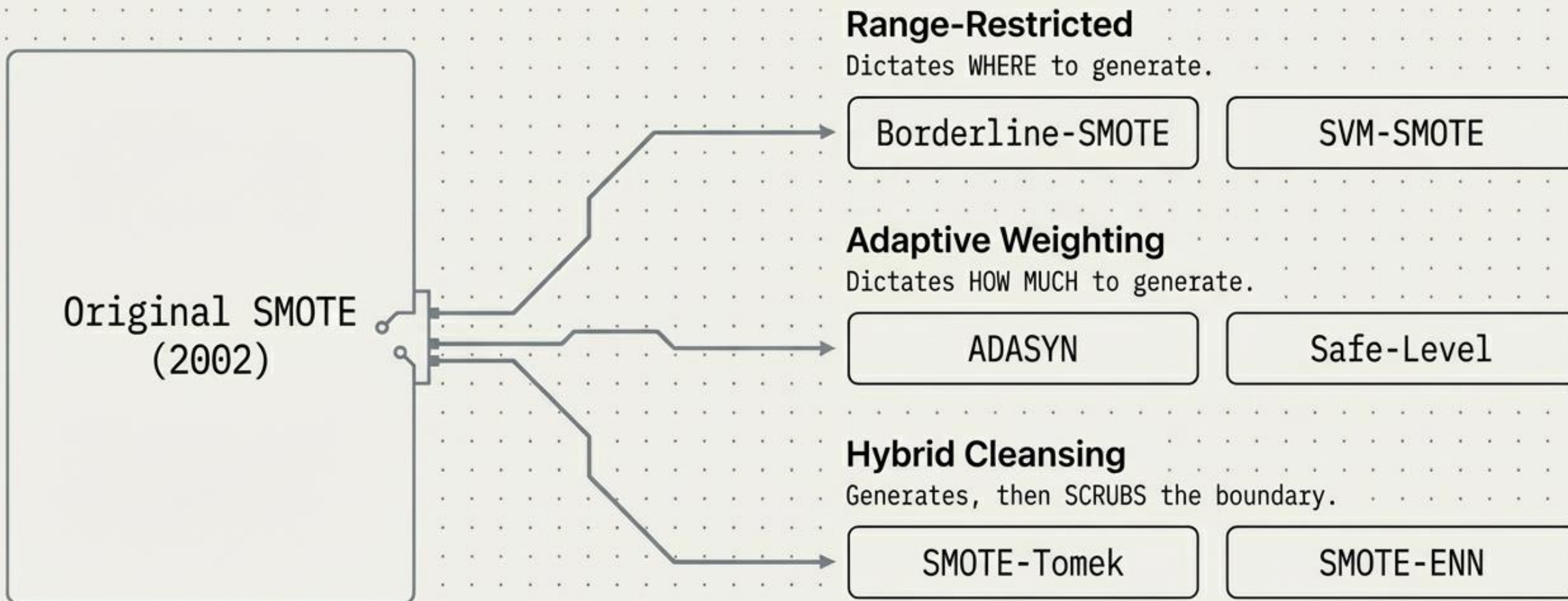
Data Leakage - CRITICAL



Applying SMOTE before the Train/Test split poisons the test set, creating a fake 99% accuracy.

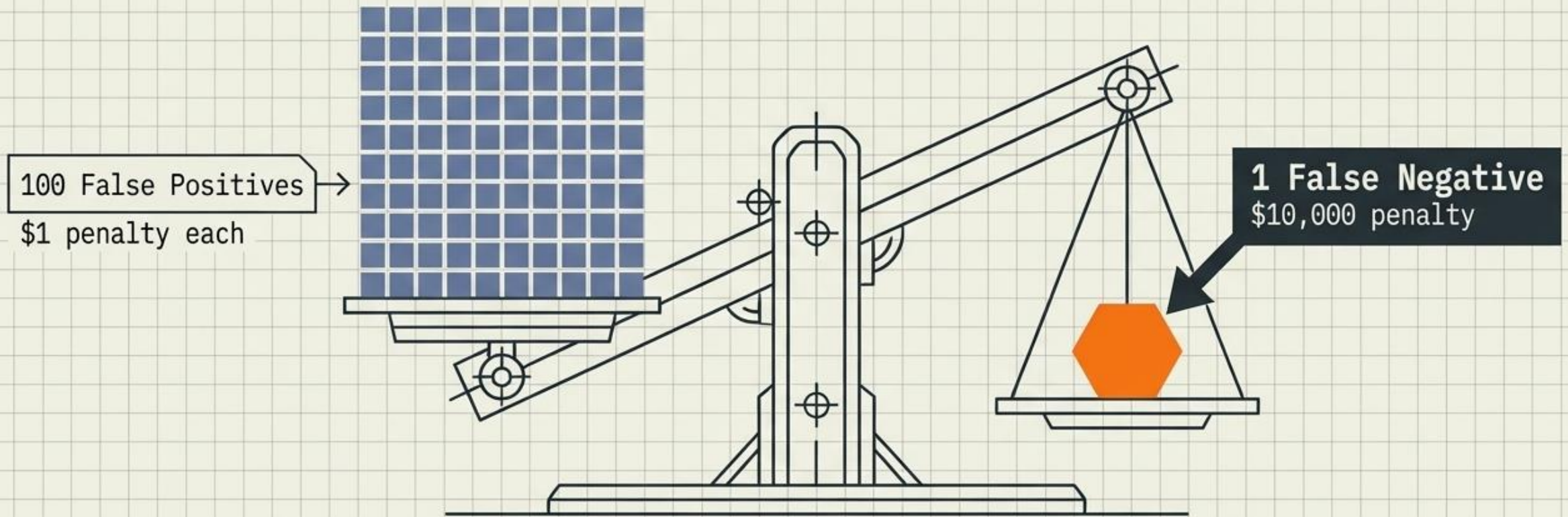
Evolution of the Arsenal

Over 85 extensions have been engineered to constrain, guide, or cleanse SMOTE's synthetic generation. They fall into three primary strategic families.



Cost-Sensitive Learning teaches the model which mistakes actually matter

Rather than generating synthetic data, we alter the objective function. We engineer the loss parameters so that missing an anomaly is mathematically more painful than raising a false alarm.



Cost sensitivity is injected via algorithmic class weights or custom matrices

Path A: Class Weights

$$\mathcal{L}(y, \hat{y}) = -[w_1 y \log(\hat{y}) + w_0 (1 - y) \log(1 - \hat{y})]$$

Objective function modified with weight multiplier.

Penalizes mistakes on the minority class disproportionately. Built natively into sklearn and XGBoost architectures.

Path B: Custom Cost Matrix

		PREDICTED CLASS	
		Negative	Positive
ACTUAL CLASS	Negative	\$0	\$1
	Positive	\$10,000	\$0

Explicitly defined penalty grids that guide custom decision thresholds for complex business domains.

How to decide class Weights?

- For binary classification:

$$w_c = \frac{N}{K \cdot N_c}$$

– Where:

- N = total number of samples
- K = number of classes (usually 2)
- N_c = number of samples in class c

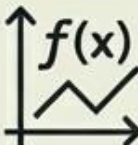
- Observation:

- Higher loss
 - But, Higher loss $\neq$ worse model
 - Weighting doesn't aim to *reduce loss value* — it aims to **change what “good” means**
 - Intuition: Unweighted: “fit most points”, Weighted: “fit important points”
- See F1-Score/Precision/Recall

The 90/10 Split Application

- Loss:
 - $\mathcal{L}_{weighted} = -[w_1 \cdot y \log(\hat{y}) + w_0 \cdot (1 - y) \log(1 - \hat{y})]$
 - Given 1,000 samples (900 of Class 0, 100 of Class 1):
 - Calculated Weights:
 - $w_0 = 0.55$ and $w_1 = 5.0$
 - The Penalty Effect:
 - If the model mispredicts Class 0, the loss is multiplied by 0.55 (Discounted).
 - If the model mispredicts Class 1, the loss is multiplied by 5.0 (Amplified).

Architectural choice: Modifying the data space vs modifying the logic

	<u>SMOTE</u>	<u>COST-SENSITIVE</u>
Intervention Layer	 Changes Data Space	 Changes Loss Logic
Leakage Risk	 High (if split improperly)	 Low
Model Compatibility	Excellent for Trees/Forests	Ideal for Linear/Neural Nets
Engineering Philosophy	Fixes the Data	Fixes the Thinking

The algorithmic workflow for triaging and correcting imbalanced data

[1. DIAGNOSE]

Is the dataset severely imbalanced? (e.g., 95:5 or 99:1)

[2. EVALUATE]

Define the exact business cost of False Positives vs. False Negatives.

[3. SELECT METRIC]

Discard Accuracy. Lock in F1, Precision, or Recall based on Step 2.

[4. INTERVENE]

Apply SMOTE (Data Layer)

or

Apply Class Weights (Loss Layer)

or a combined architecture.

System rules for engineering robust imbalanced models

RULE_01

Accuracy is a Trap.

Always rely on the Confusion Matrix to reveal the true structural health of the model.

RULE_02

Metrics are Business Rules.

Map math directly to real-world costs (Precision vs. Recall).

RULE_03

Beware of Leakage.

Never oversample the feature space before executing the Train/Test split.

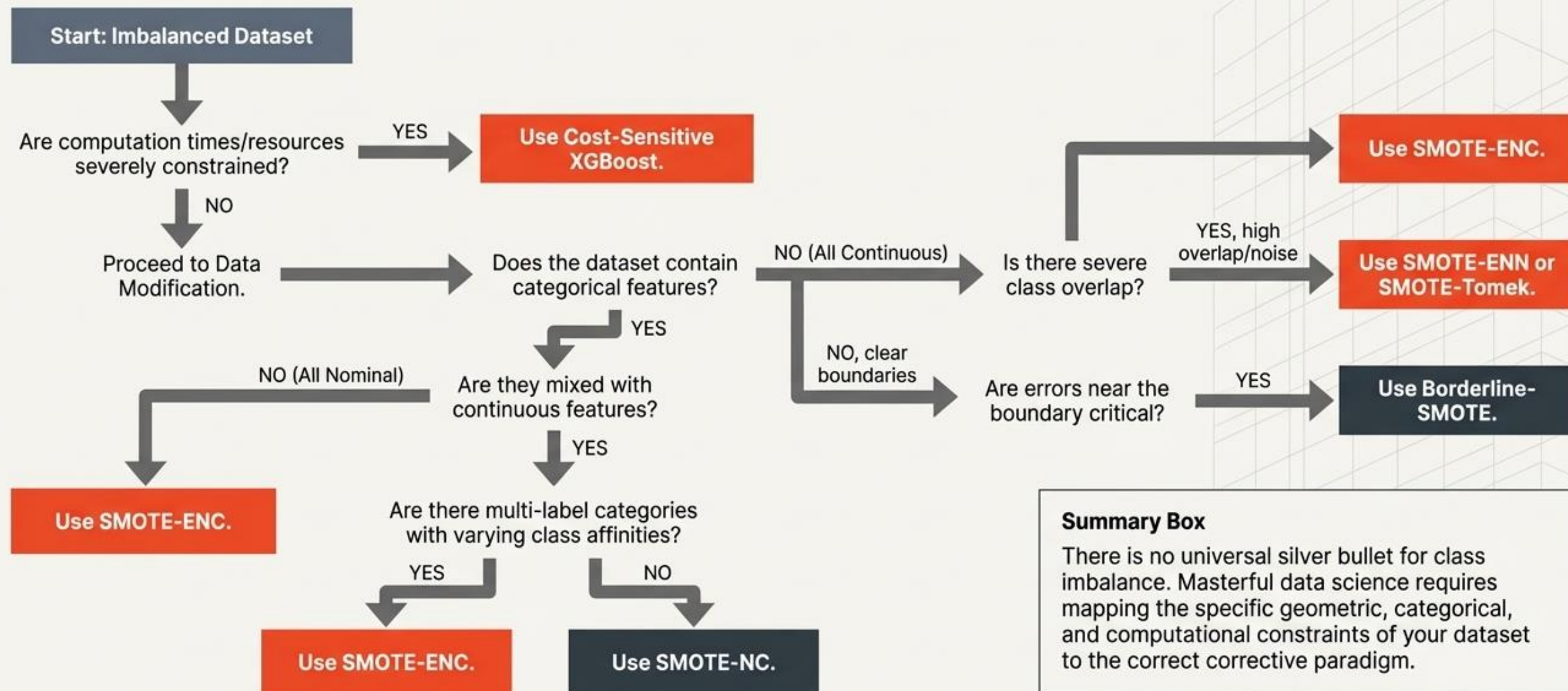
RULE_04

Balance the System.

Fix the underlying data distribution (SMOTE) or fix the algorithmic penalty (Cost-Sensitive).

Synthesis: The Practitioner's Decision Framework

The Imbalanced Learning Selection Guide



Summary Box

There is no universal silver bullet for class imbalance. Masterful data science requires mapping the specific geometric, categorical, and computational constraints of your dataset to the correct corrective paradigm.